

FIT AR Navigation App (FITARNA)

Vincenzo Barager, Dathan Dixon, Jacob Hall-Burns, Ethan Wadley

Faculty Advisor: Eraldo Ribeiro, Dept. of Computer Science, Florida Institute of Technology

Motivation

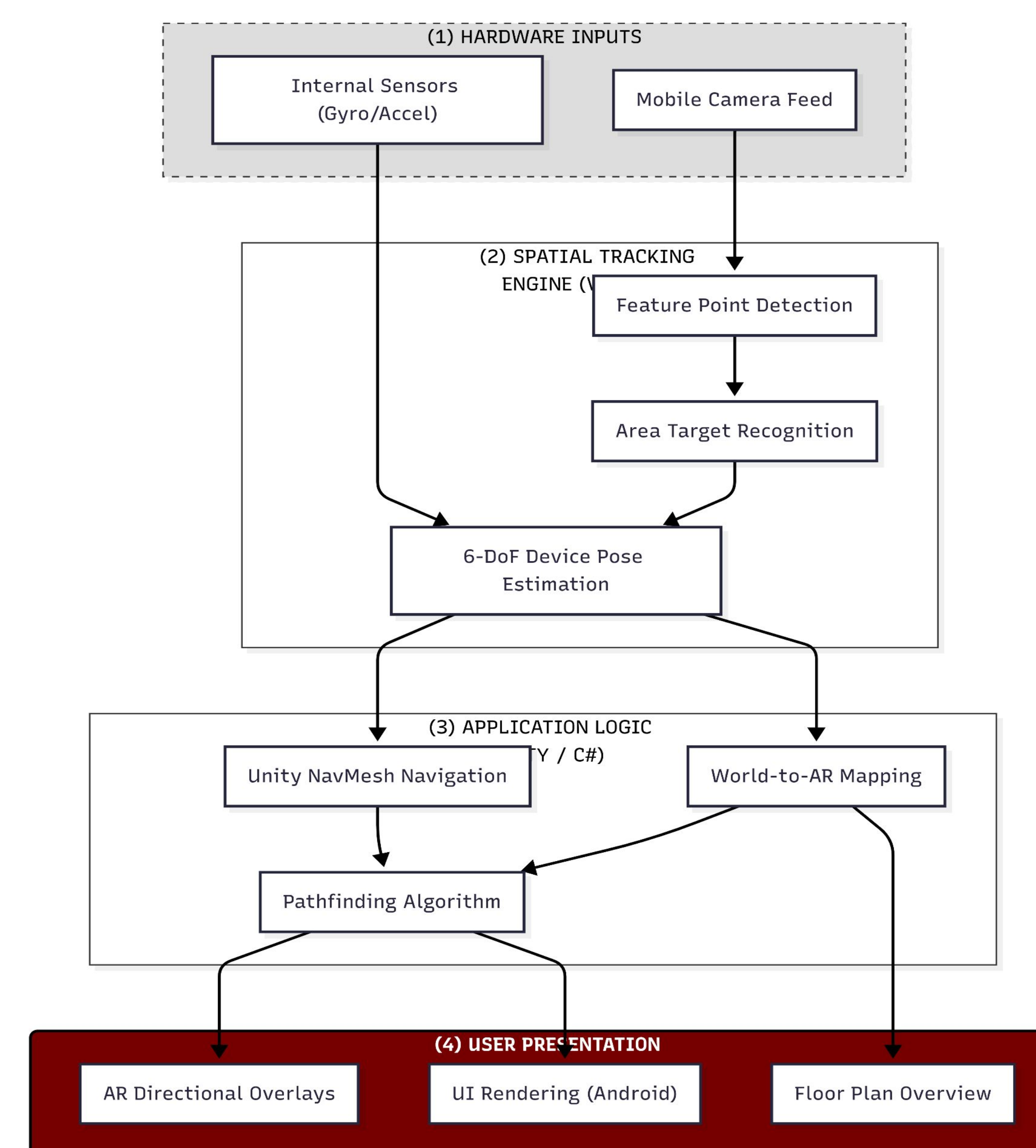
- **Navigation Complexity:** Large-scale academic facilities like the Evans Library present significant wayfinding challenges for new students and visitors.
- **GPS Limitations:** Standard satellite-based navigation fails indoors due to signal attenuation and lack of floor-level granularity.
- **The Solution:** FITARNA leverages Augmented Reality to provide intuitive, real-time visual guidance, bridging the gap between digital maps and the physical environment.



Implementation and Approach

- **Environment Mapping:** Conducted extensive spatial scans of the Evans Library using Vuforia Area Targets to create a high-fidelity digital twin for persistent tracking.
- **Hybrid AR Framework:** Integrated Unity's AR Foundation with the Vuforia Engine to leverage cross-platform ARCore/ARKit features while maintaining superior object and area recognition.
- **Pathfinding Logic:** Implemented a custom NavMesh within Unity to calculate real-time, shortest-path routes between navigational waypoints.
- **Spatial UI Overlay:** Developed a user interface that projects 3D directional markers and "breadcrumbs" onto the physical world, ensuring intuitive wayfinding in a complex indoor environment.

System Architecture



Project Goals

- **High-Precision Localization:** Achieve sub-meter indoor positioning accuracy without the use of GPS by leveraging AR spatial mapping.
- **Intuitive AR Wayfinding:** Develop a seamless navigation experience using real-time 3D directional overlays.
- **Dynamic Point-of-Interest System:** Implement a searchable database of campus locations, such as specific library wings, study rooms, and help desks.

Software Stack

